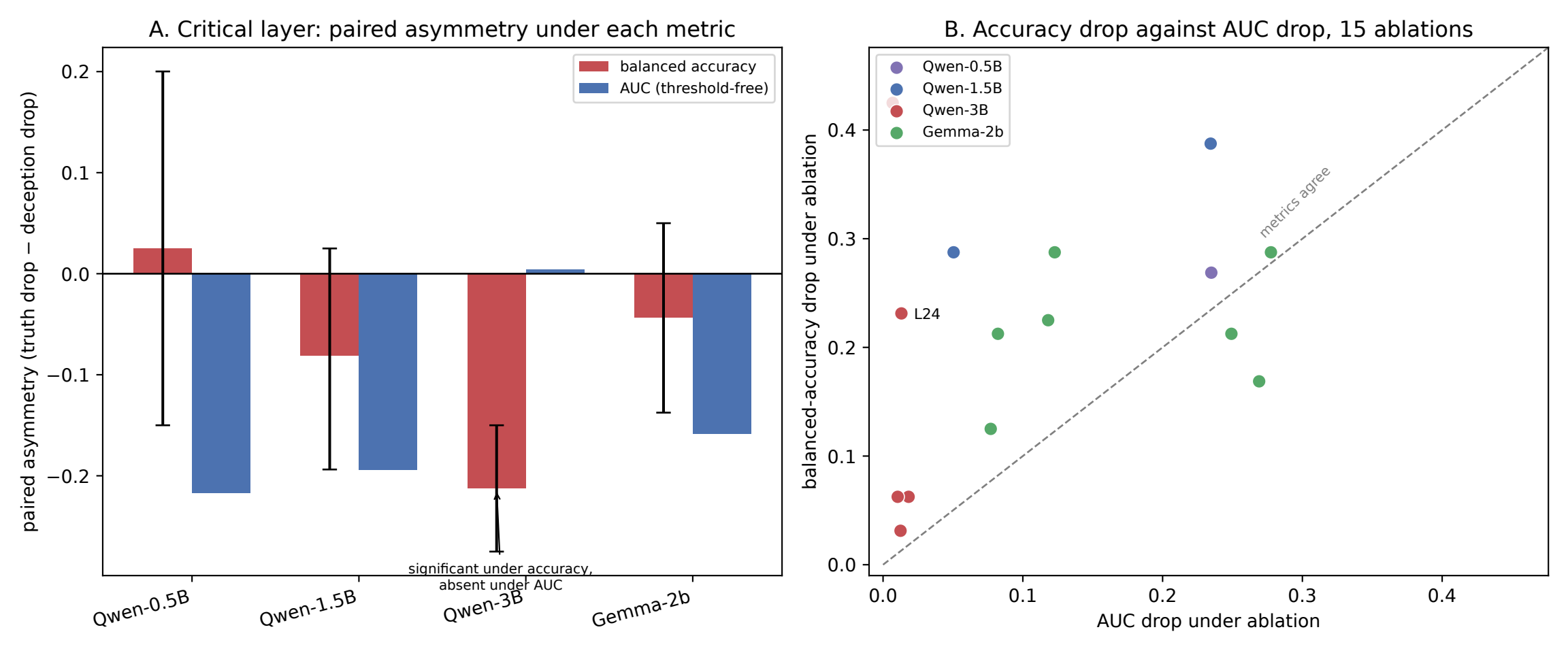

A. Critical layer: paired asymmetry under each metric
balanced accuracy
AUC (threshold-free)
paired asymmetry (truth drop − deception drop)
0.2
0.1
0.0
−0.1
−0.2
significant under accuracy,
absent under AUC
Qwen-0.5B
Qwen-1.5B
Qwen-3B
Gemma-2b
B. Accuracy drop against AUC drop, 15 ablations
Qwen-0.5B
Qwen-1.5B
Qwen-3B
Gemma-2b
metrics agree
L24
balanced-accuracy drop under ablation
AUC drop under ablation
0.0
0.1
0.2
0.3
0.4